

Polygon Material IDs

The Polygon: Material IDs rollout has controls for applying material IDs to the object and smoothing the object on a per-polygon basis. Material IDs allow multiple materials to be applied to a single object. Materials in a



Torus knot with a particular material applied

Multi/Sub-Object material will be applied to the polygons containing the corresponding ID. Smoothing groups control how the renderer calculates smoothing between the polygons, resulting in a smooth or faceted look.

Modifying Sub-Objects

Standard 3ds Max modifiers can be applied to sub-objects so that they affect only part of an object. An FFD, for example, can be applied to the nose of a character to assist in reshaping it.

To apply a modifier to a sub-object selection, the sub-object must be active when the modifier is applied. The sub-object can be selected by using an Edit Poly, Edit Mesh, or Edit Patch modifier. After the modifier is applied, only the selection will be affected. If needed, you can change the selection or add a falloff by using soft selection to fine-tune the process.

If selection is all that's needed, a Select Poly, Select Mesh, or Select Patch modifier will perform the same task with less overhead.

2.9 Using Subdivision Surfaces

Subdivision surfaces combine the ease of polygonal modelling with the smoothness of a patch surface, giving you the best of both worlds. Subdivision surfaces are used extensively in character animation because they can define almost any complex smooth surface while keeping the underlying geometry light enough to be deformed quickly.

3ds Max has several ways to create a subdivision surface. The MeshSmooth and TurboSmooth modifiers can be applied to any editable poly or editable mesh object. Editable poly objects also have a subdivision surface built into the base object.